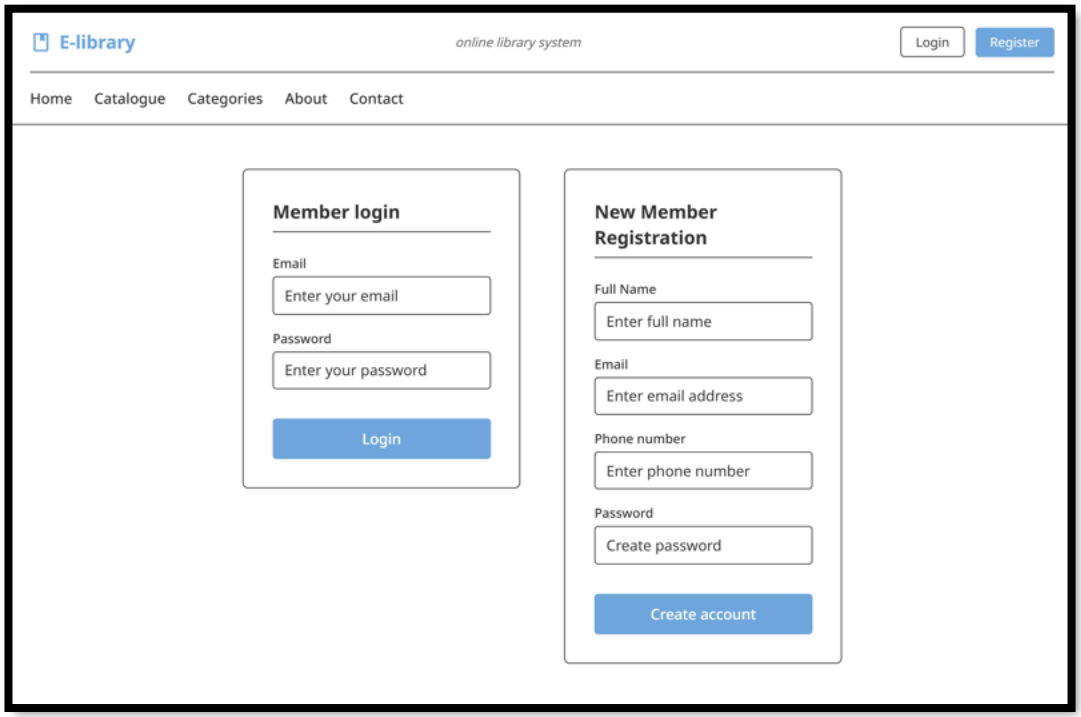
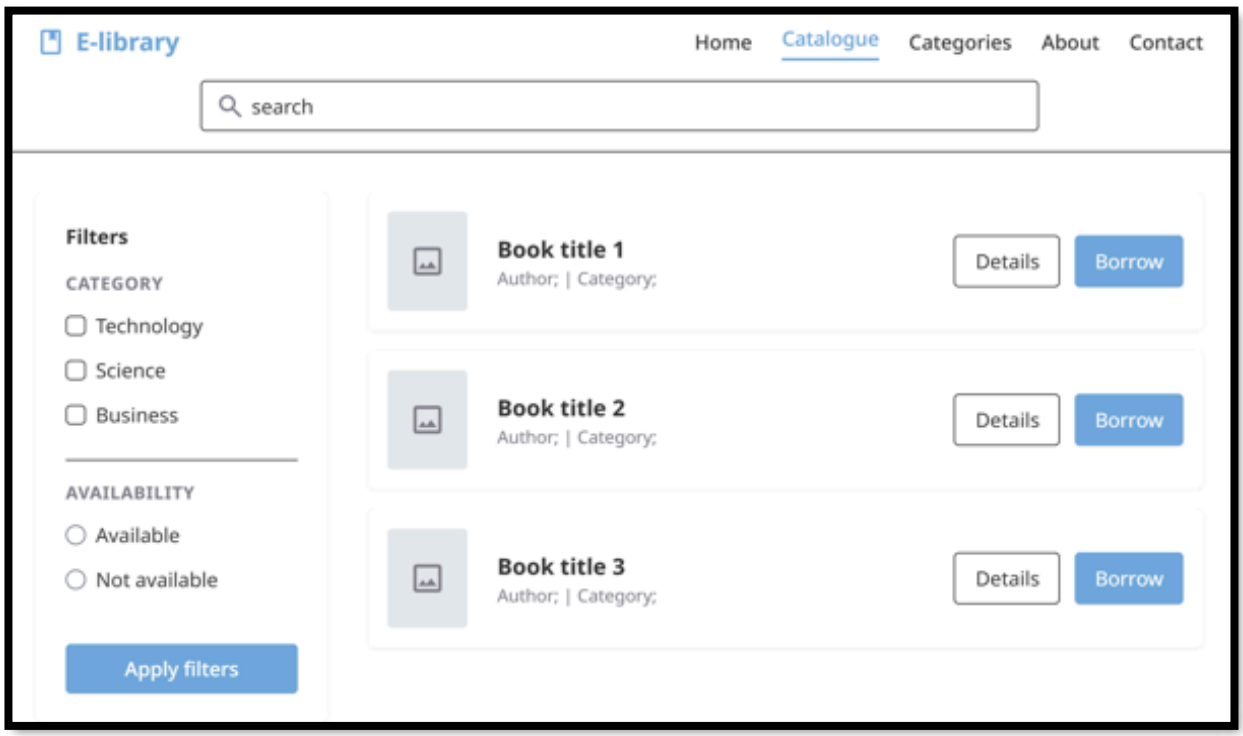
Front-end design

Login and registration



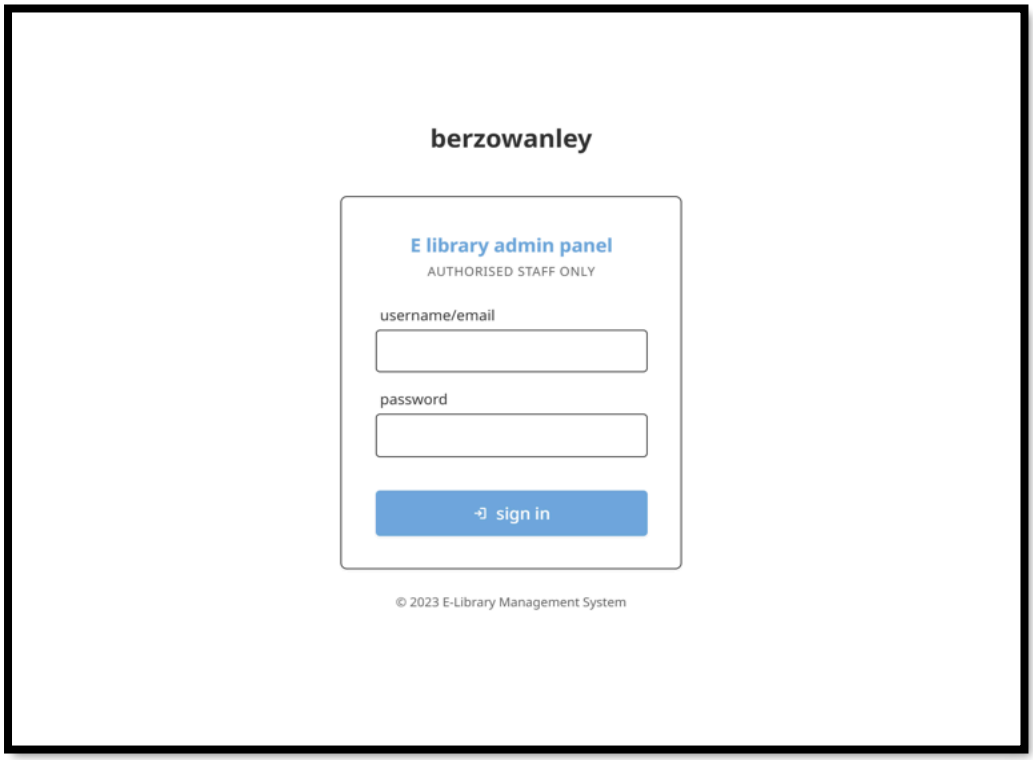
Book Selection



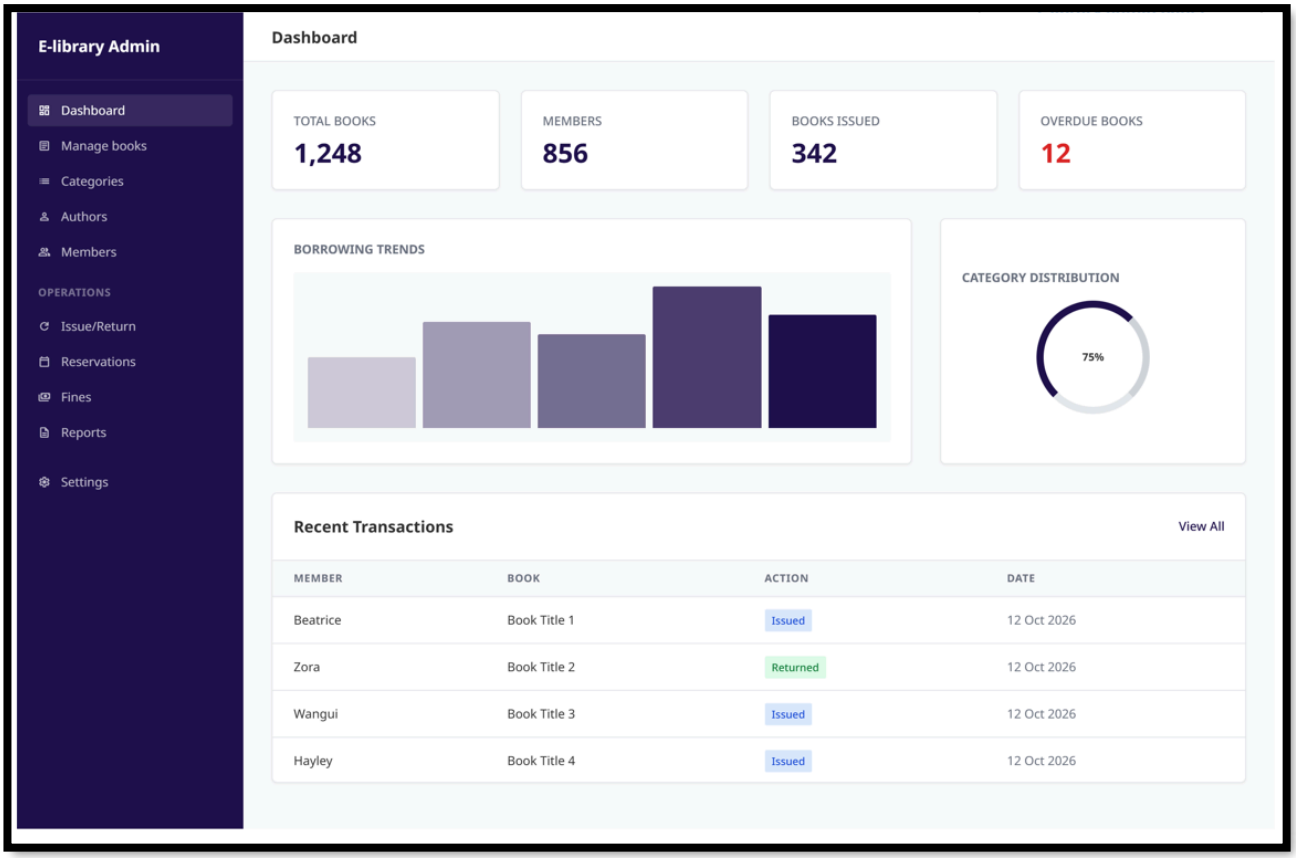
2. Admin panel design

Edit with WPS Office

Admin login

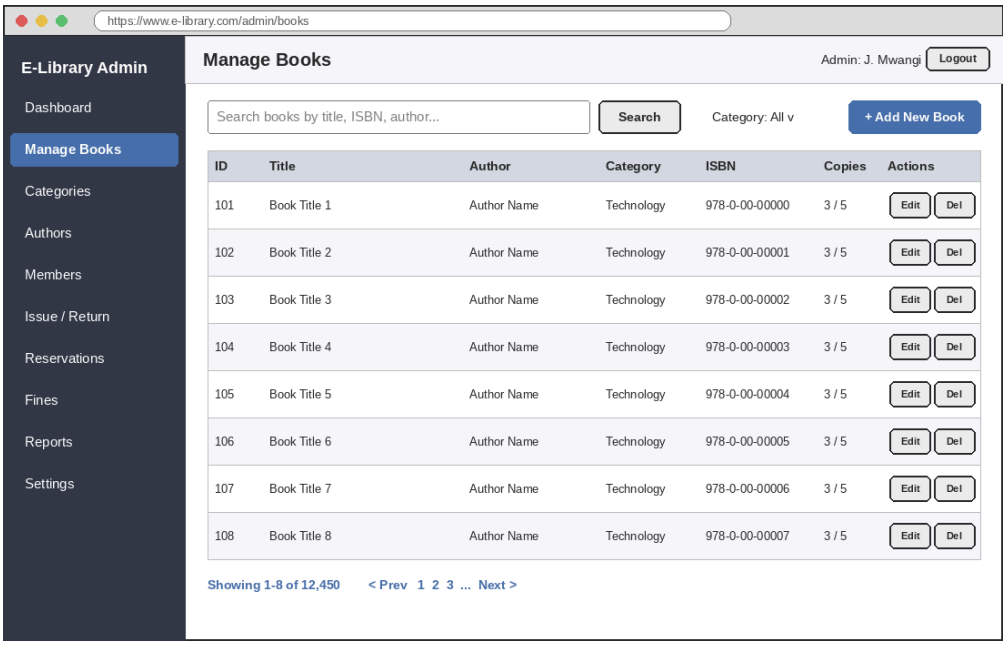


Dashboard

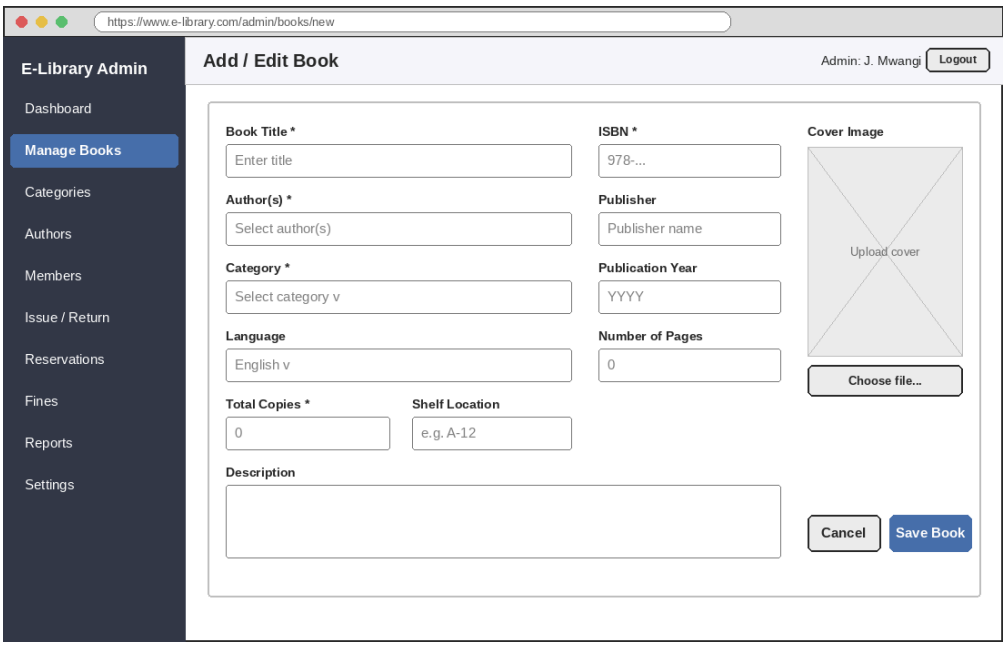


3.3 Manage books

Edit with WPS Office

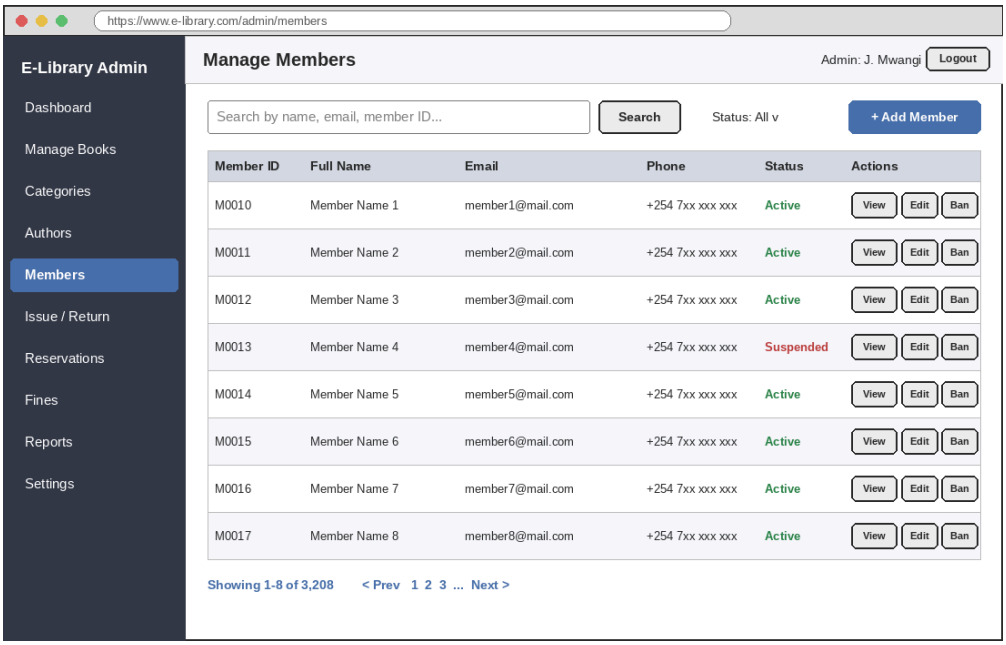


3.4 Add or edit a book

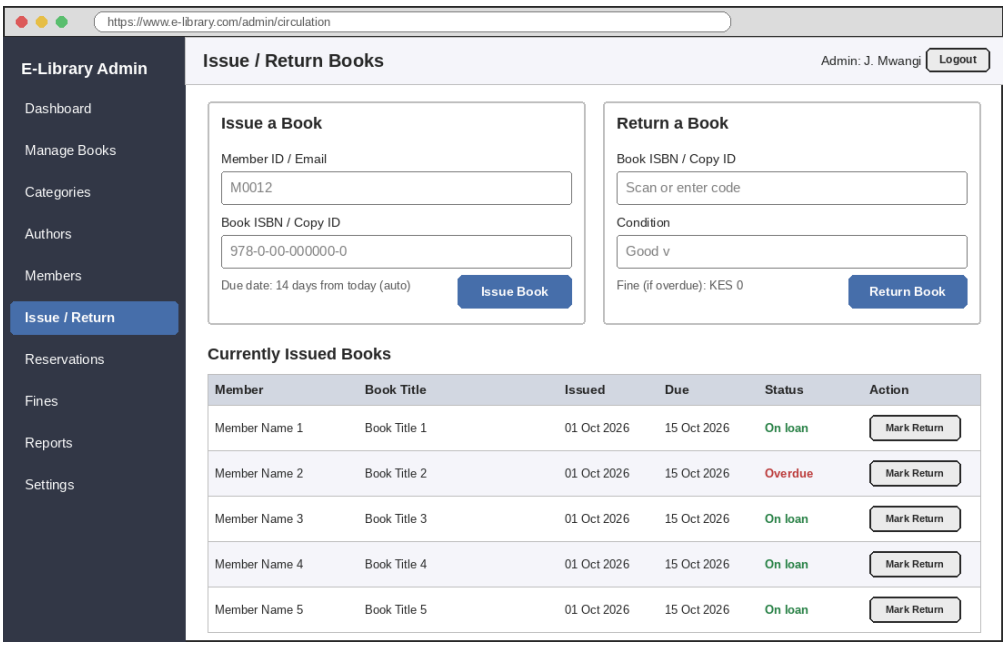


3.5 Manage members

Edit with WPS Office

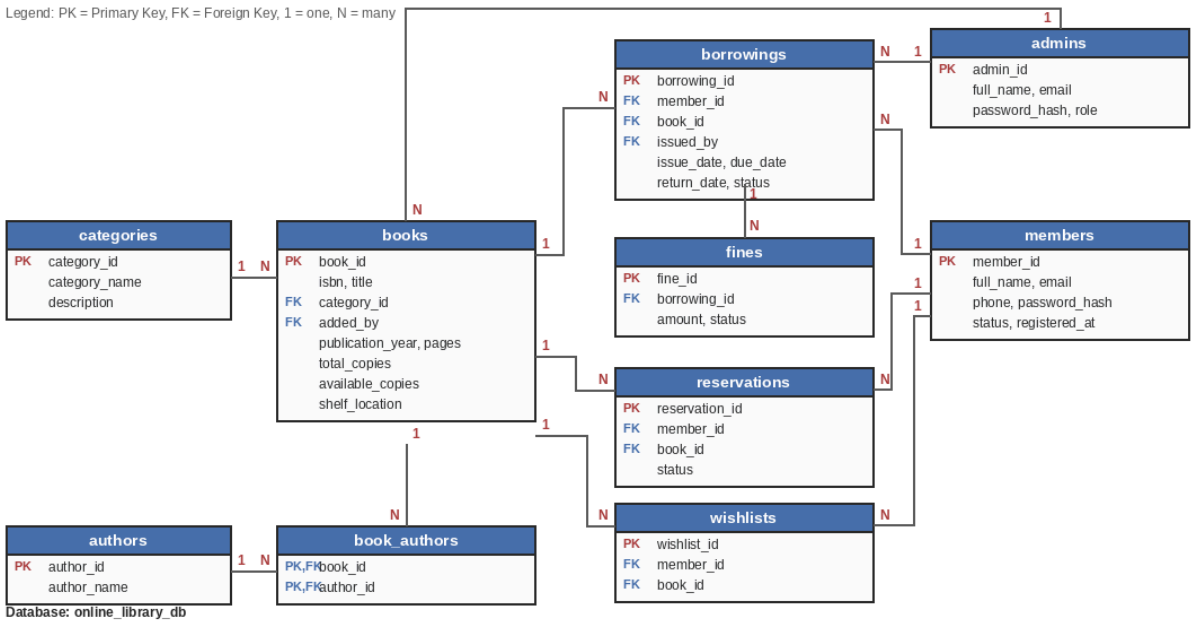


3.6 Issue and return books



Edit with WPS Office

4. Database schema



admins

Column	Data type	Key and constraints
admin_id	INT	PK, AUTO_INCREMENT
full_name	VARCHAR(100)	NOT NULL
=		
email	VARCHAR(100)	NOT NULL, UNIQUE
password_hash	VARCHAR(255)	NOT NULL
role	ENUM	'super_admin' or 'librarian', NOT NULL, DEFAULT 'librarian'
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP

members

Column	Data type	Key and constraints
member_id	INT	PK, AUTO_INCREMENT
full_name	VARCHAR(100)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
phone	VARCHAR(20)	Can be empty
password_hash	VARCHAR(255)	NOT NULL
status	ENUM	'active', 'suspended' or 'inactive', NOT NULL, DEFAULT 'active'
registered_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP

categories

Column	Data type	Key and constraints
category_id	INT	PK, AUTO_INCREMENT
category_name	VARCHAR(80)	NOT NULL, UNIQUE
description	TEXT	Can be empty

authors

Column	Data type	Key and constraints
author_id	INT	PK, AUTO_INCREMENT
author_name	VARCHAR(100)	NOT NULL
biography	TEXT	Can be empty

Edit with WPS Office

books

Column	Data type	Key and constraints
book_id	INT	PK, AUTO_INCREMENT
isbn	VARCHAR(20)	NOT NULL, UNIQUE
title	VARCHAR(200)	NOT NULL
category_id	INT	FK, NOT NULL, references categories(category_id), ON DELETE RESTRICT
publisher	VARCHAR(100)	Can be empty
publication_year	SMALLINT	CHECK (publication_year BETWEEN 1000 AND 2100)
language	VARCHAR(30)	NOT NULL, DEFAULT 'English'
pages	INT	CHECK (pages > 0)
total_copies	INT	NOT NULL, DEFAULT 1, CHECK (total_copies >= 0)
available_copies	INT	NOT NULL, CHECK (available_copies >= 0 AND available_copies <= total_copies)
shelf_location	VARCHAR(20)	Can be empty
description	TEXT	Can be empty
cover_image	VARCHAR(255)	File path or URL, can be empty
added_by	INT	FK, references admins(admin_id), ON DELETE SET NULL
created_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP

book_authors

Column	Data type	Key and constraints
book_id	INT	PK, FK, references books(book_id), ON DELETE CASCADE
author_id	INT	PK, FK, references authors(author_id), ON DELETE CASCADE

borrowings

Column	Data type	Key and constraints
borrowing_id	INT	PK, AUTO_INCREMENT
member_id	INT	FK, NOT NULL, references members(member_id)
book_id	INT	FK, NOT NULL, references books(book_id)
issued_by	INT	FK, references admins(admin_id), ON DELETE SET NULL
issue_date	DATE	NOT NULL, DEFAULT CURRENT_DATE
due_date	DATE	NOT NULL, CHECK (due_date >= issue_date)
return_date	DATE	Empty untilbook comes back
status	ENUM	'borrowed', 'returned' or 'overdue', NOT NULL, DEFAULT 'borrowed'

reservations

Column	Data type	Key and constraints
reservation_id	INT	PK, AUTO_INCREMENT
member_id	INT	FK, NOT NULL, references members(member_id), ON DELETE CASCADE
book_id	INT	FK, NOT NULL, references books(book_id), ON DELETE CASCADE
reservation_date	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
status	ENUM	'pending', 'fulfilled', 'cancelled' or 'expired', NOT NULL, DEFAULT 'pending'

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fines

Column	Data type	Key and constraints
fine_id	INT	PK, AUTO_INCREMENT
borrowing_id	INT	FK, NOT NULL, UNIQUE, references borrowings(borrowing_id)
amount	DECIMAL(8,2)	NOT NULL, CHECK (amount >= 0)
reason	VARCHAR(100)	NOT NULL, DEFAULT 'Overdue'
status	ENUM	'unpaid' or 'paid', NOT NULL, DEFAULT 'unpaid'
issued_date	DATE	NOT NULL, DEFAULT CURRENT_DATE
paid_date	DATE	Empty until the fine is paid

wishlists

Column	Data type	Key and constraints
wishlist_id	INT	PK, AUTO_INCREMENT
member_id	INT	FK, NOT NULL, references members(member_id), ON DELETE CASCADE
book_id	INT	FK, NOT NULL, references books(book_id), ON DELETE CASCADE
added_at	DATETIME	NOT NULL, DEFAULT CURRENT_TIMESTAMP
(member_id, book_id)		UNIQUE together, so a book can only be wishlisted once per member

Edit with WPS Office